

A HAPPY MOMENT
Chandana and Daka with VIPs in ISRO's mission control room at Vikram S's launch

"Private players with ISRO collaboration will help build an ecosystem for the sector to grow"

– S. SOMANATH
Chairman, ISRO

"The path-breaking initiatives were designed to keep the doors open for the private industry in the space sector, particularly start-ups"

– JITENDRA SINGH
MoS for Space & Atomic Energy

"The aim is to get a much bigger share of the $440 billion global space market"

– PAWAN GOENKA
Chairperson, IN-SPACe

attained hypersonic speed, exceeding Mach5, or five times the speed of sound. Chandana and the others watched the progress of the rocket on giant computer screens and clapped as it followed the projected trajectory. Having reached the desired sub-orbital height of 89.5 km, it began its descent, and crashed into the Bay of Bengal 115 km away from the launch pad. "Bullseye," Chandana yelled even as those in the gallery cheered loudly. Skyroot had made space history of sorts in India.

For Pawan Goenka, IN-SPACe chairperson and former managing director of Mahindra & Mahindra, Skyroot's success in the first attempt should convince the sceptics that India's private sector has what it takes to do well in space. "I admit this is a baby step," Goenka said, "but it is a very good beginning for the long journey ahead." It also gave one a sense of how ISRO and the private sector would work together. Goenka was encouraged by the fact that there are over 101 start-ups that have registered after the government's new policy. Of these, over a quarter are at an advanced stage of putting out their first products. Among them is AgniKul Cosmos that was incubated in IIT-Madras in 2018. The Chennai-based start-up is expected to test its launch vehicle called Agnibaan (arrow of fire) next month.

ISRO's Somanath believes that with start-ups like Skyroot beginning to make a mark, private investors are also beginning to exhibit confidence in them. The efforts of Skyroot and AgniKul got a boost when private investors backed them with serious cash. Skyroot was able to raise Rs 500 crore and AgniKul half that amount so far. Skyroot is now gearing up to launch Vikram 1, an orbital flight, and will be offering its services to satellite manufacturers and buyers who are looking for speedy and cost-effective launchers.

Somanath believes the two-pronged approach of encouraging start-ups on the one hand and opening the doors to big private players to collaborate with ISRO on the other will create the ecosystem required for the space sector to grow more rapidly than before. It will leave ISRO to focus more on research and development in the field of advanced space technology while continuing with its existing missions to meet national requirements. This includes *Gaganyaan*, its most ambitious project date, to send three Indian astronauts into the orbit around the Earth. India will then join the big boys of space sooner rather than later or never. ■